

COINDESK RESEARCH NOTE

Ethereum On-chain Transaction Volume: Layered Information

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INTRODUCTION

On-chain transaction volumes seem straightforward on the surface, especially when they are available for all to see on public blockchains. You add up the number of tokens that move between accounts in a given day – simple, right?

On Ethereum, however, on-chain transaction volumes (herein known as "transaction volumes") represent the movement of much more than ether, Ethereum's native token. Tracking them gives an idea of the health of the network. It also highlights the complex layers of activity powered by Ethereum, and allows a glimpse of the network's potential and reach.

In this Research Note, we're going to look at what information Ethereum transactions contain, what their trends say about the evolution of the Ethereum blockchain, as well as what they don't say. We will also look at the volumes for ether as well as different Ethereum-based decentralized finance (DeFi) applications.

As a convention, we capitalize the names of blockchains (such as Ethereum and Bitcoin) and use lowercase for the names of the tokens (such as ether and bitcoin). All dollars refer to U.S. dollars (USD).

WHAT DO ETHEREUM TRANSACTIONS REPRESENT?

As with Bitcoin, a transaction on the Ethereum blockchain passes value and/or information from one account to another. If, for example, Bob wants to send 1 ETH to Alice, Bob will initiate a transaction that asks the Ethereum protocol to validate his holdings, transfer 1 ETH from his address to that specified, and then update the network state.

There the similarities end, however.

On Ethereum, most transactions are not between two people, but between people and snippets of code, or even between two different snippets of code.

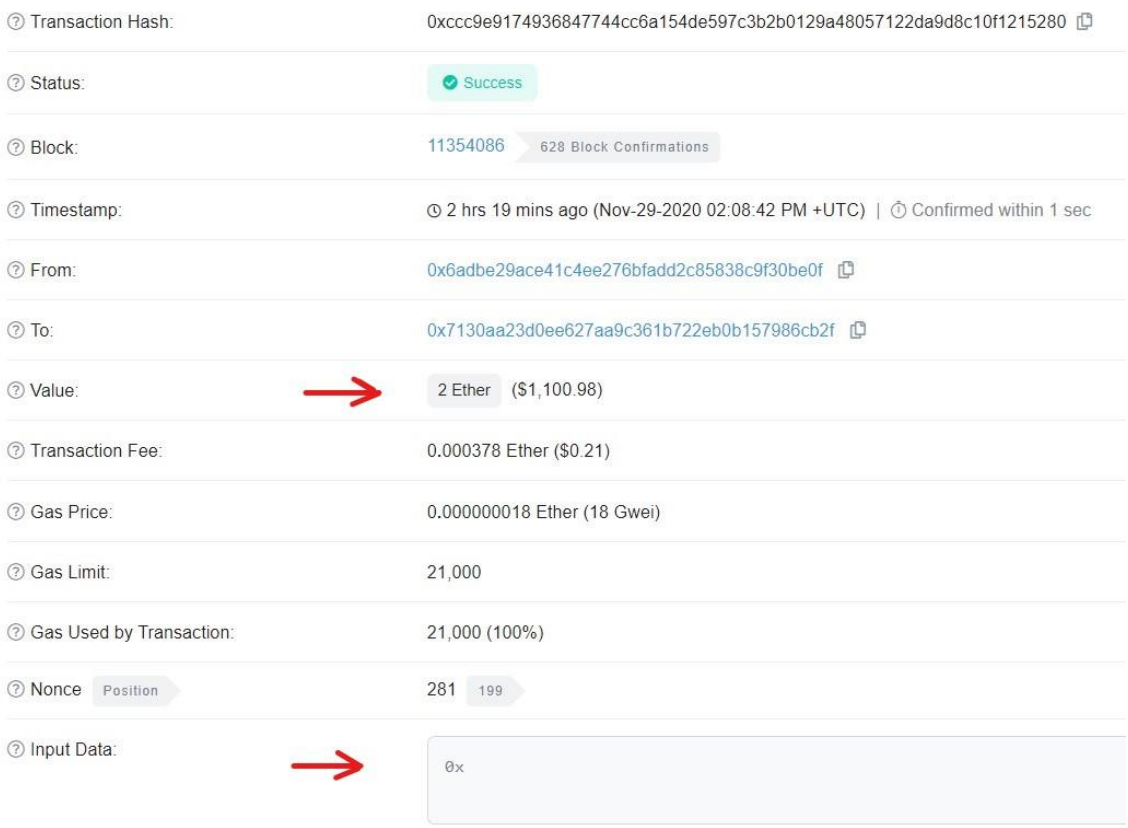
One of Ethereum's main value propositions as a distributed blockchain is that it can run decentralized applications (dapps) across the network. Dapps are a combination of a user interface and one or several "smart contracts," or scripts that execute a function and produce an outcome in response to an input.

These inputs can be tokens, or "value." They can also be instructions, or "data."

These inputs constitute the main components of any transaction on Ethereum. If the transaction contains a non-zero number of ETH, it is a transfer of value, or a payment. It can also contain data, which means that it is likely to be a payment to a smart contract. This data is written on to the Ethereum protocol explicitly for the designated smart contract. (Some transactions contain just data, which makes it more like a dapp instruction, such as a transfer between or within dapps. As mentioned above, dapps are a combination of one or several smart contracts.)

Transactions that just contain value and no instruction data are usually transfers of ETH between holders (such as from an investor to an exchange, or as a payment or gift). They can also be the result of a transfer between two different addresses belonging to the same entity – this happens when people change custody providers, or when an exchange moves holdings for security purposes.

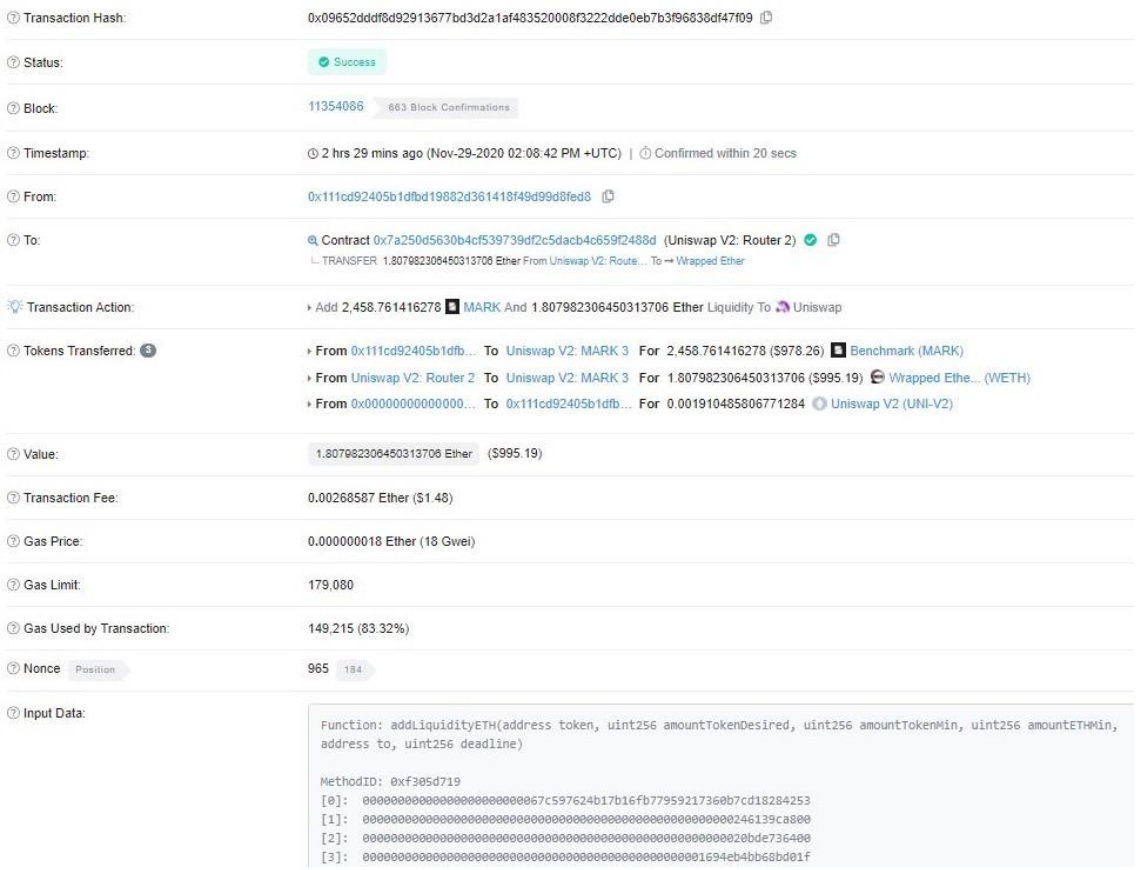
What a transfer of ETH between holders looks like



(Source: Etherscan.io)

When the transaction also contains input data, it usually represents the transfer of ETH to a dapp, such as this example which sends some ETH to decentralized exchange Uniswap.

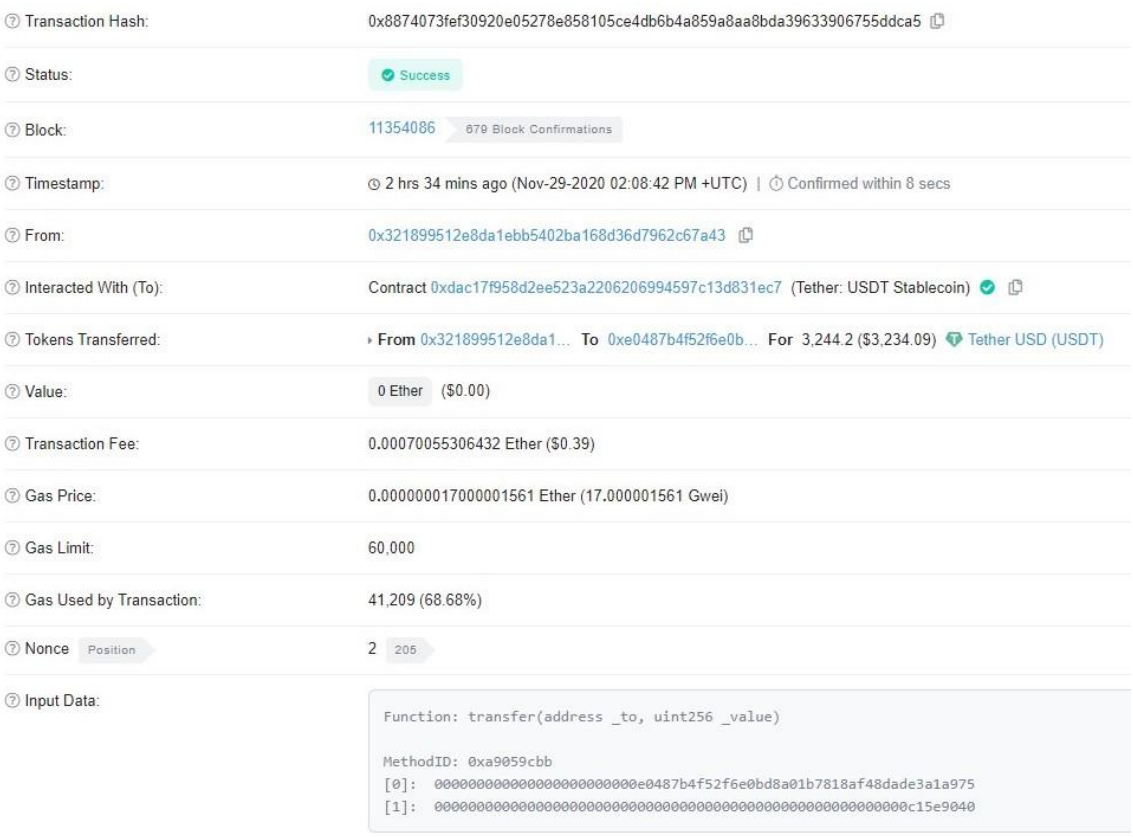
What a transfer of ETH to a dapp looks like



(Source: Etherscan.io)

Many transactions do not transfer ETH at all, but they do transfer other Ethereum-based assets between accounts, such as this example which shows a transfer of stablecoin tether from one Ethereum address to another.

What a transfer of an Ethereum-based token looks like



(Source: Etherscan.io)

Unlike relatively simple Bitcoin transactions, which indicate a transfer of value from one address to another, Ethereum transactions represent a much broader range of economic activity than that represented by the transfer of ETH.

This is not just due to the existence of a wide array of smart contracts that connect with the Ethereum blockchain; it is also due to the range of unit types that could be involved.

UNIT TYPES

While transactions on the Bitcoin blockchain can only handle one type of unit – bitcoin – Ethereum transactions can reference a range of unit types, of which ETH is one.

All the unit types are constructed according to certain standards, but they can have differing characteristics and value.

The variety is vast, and broadening, but the types of tokens can be broken down into categories according to the "standards" on which they were built.

To facilitate the creation of different types of tokens while ensuring interoperability with most Ethereum-based dapps, developers devised a series of technical standards, or templates.

The most common is the ERC20 standard, which allows almost anything to be represented by an Ethereum-based token which can be fractionalized and traded. Etherscan currently tracks almost 340,000 ERC20 tokens at time of writing. Among the more popular ones are stablecoins tether (USDT) and USDC, smart contract oracle Chainlink's LINK and crypto exchange Binance's BNB.

Some dapps such as decentralized exchange Uniswap allow users to transfer ETH to a Uniswap account in order to convert it into an ERC20 representation of ETH (Wrapped Ether), which has more compatibility with many dapps.

Another popular template is the ERC721, which configures non-fungible tokens used to create digital art and representations of real-word assets. The ERC721 tokens with the highest volume are at time of writing Sorare, a fantasy football game, and collectible cat characters CryptoKitties. According to Etherscan, at time of writing over 7,600 tokens use the ERC721 standard.

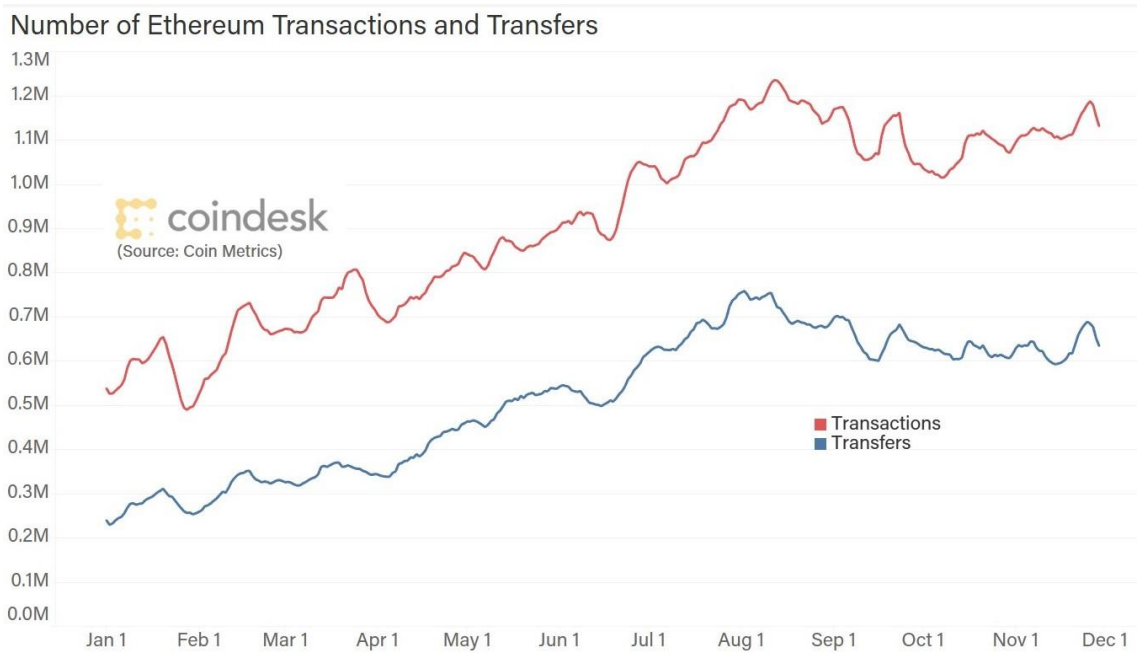
TRANSACTION AND TRANSFER VOLUMES

Parsing all the transactions on the Ethereum blockchain gives us insight into the activity growth in various areas of the ecosystem.

The growth in the number of transactions is a useful indicator of overall ecosystem activity. As we saw above, not all Ethereum transactions involve ETH, and so this metric is not directly related to demand for ETH. But it does point to the blockchain's utility, and since ETH is the native token, growth in Ethereum activity is an indicator of the growing value of ETH.

The number of transfers of ETH tells a more relevant, but incomplete, story. This refers to the number of transactions on Ethereum that transfer ETH from one address to another.

Ethereum transactions don't always result in ETH transfers

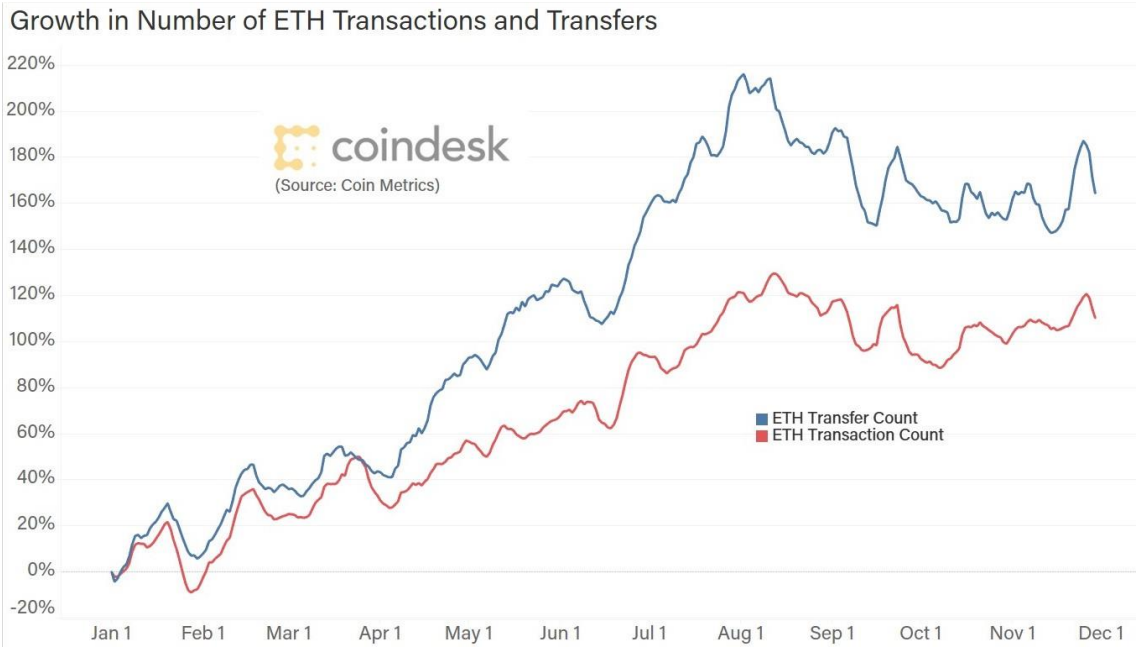


(NOTE: An important difference between the Bitcoin and the Ethereum blockchains is the difference between transactions and transfers. With Bitcoin, the number of transfers is much higher than the number of transactions. This is because each transaction can contain multiple transfers of BTC.

Ethereum transactions can also contain multiple transfers of ETH, but it's more likely that they will contain no transfers at all. Most ETH transactions are contract to contract, with no ETH changing hands (other than fees). So, in Ethereum, the number of transactions is much greater than the number of transfers of ETH.)

Looking at the relative growth rates of Ethereum transactions and transfers, we see that, over 2020, the number of transfers of ETH has grown faster than the number of Ethereum transactions.

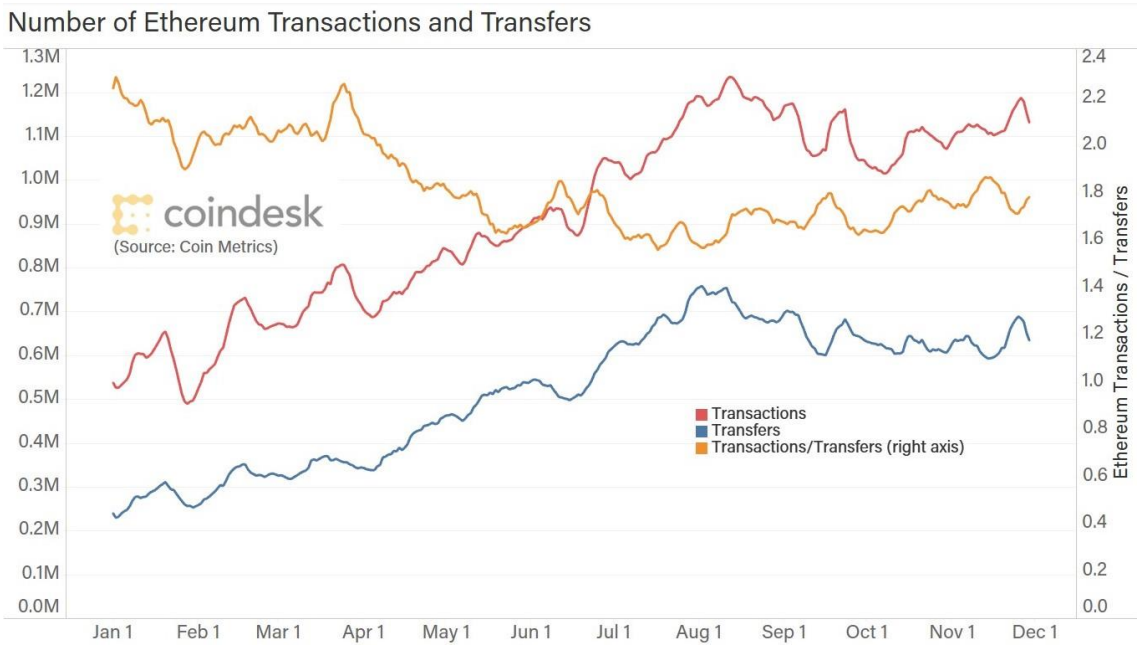
The number of ETH transfers is growing faster than transactions...



This is in spite of the strong growth in DeFi activity during the year, and shows that Ethereum growth is not just about new token types and decentralized applications – it is also reflected in trading and utility of the native token ETH.

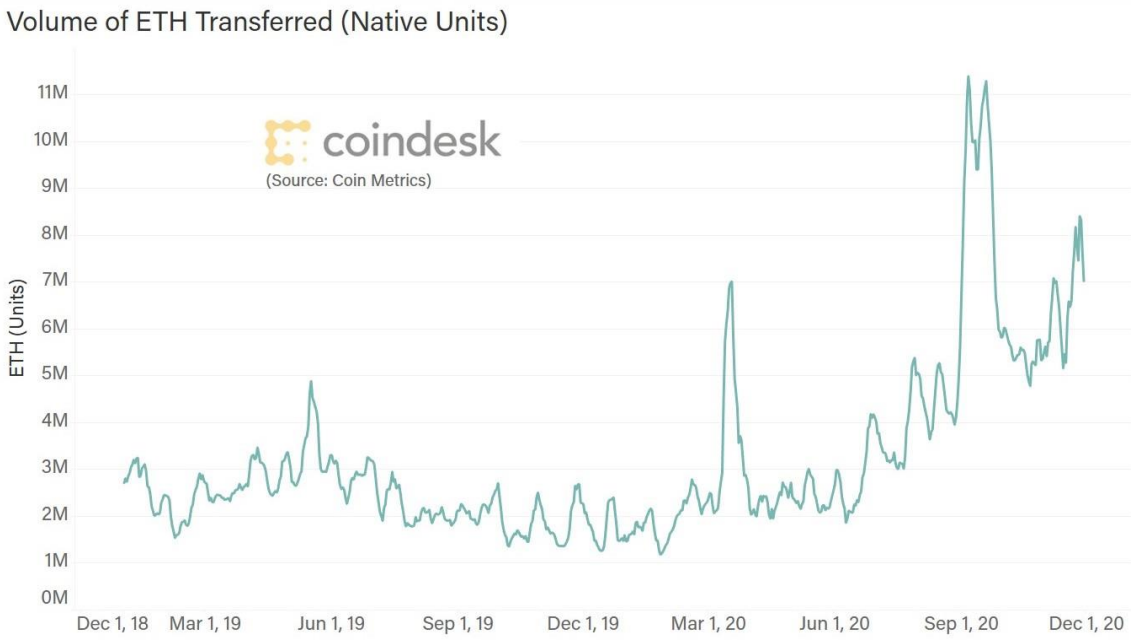
This can also be seen in the decline over the year in the ratio between transactions and transfers:

...which hints at strong interest in ETH



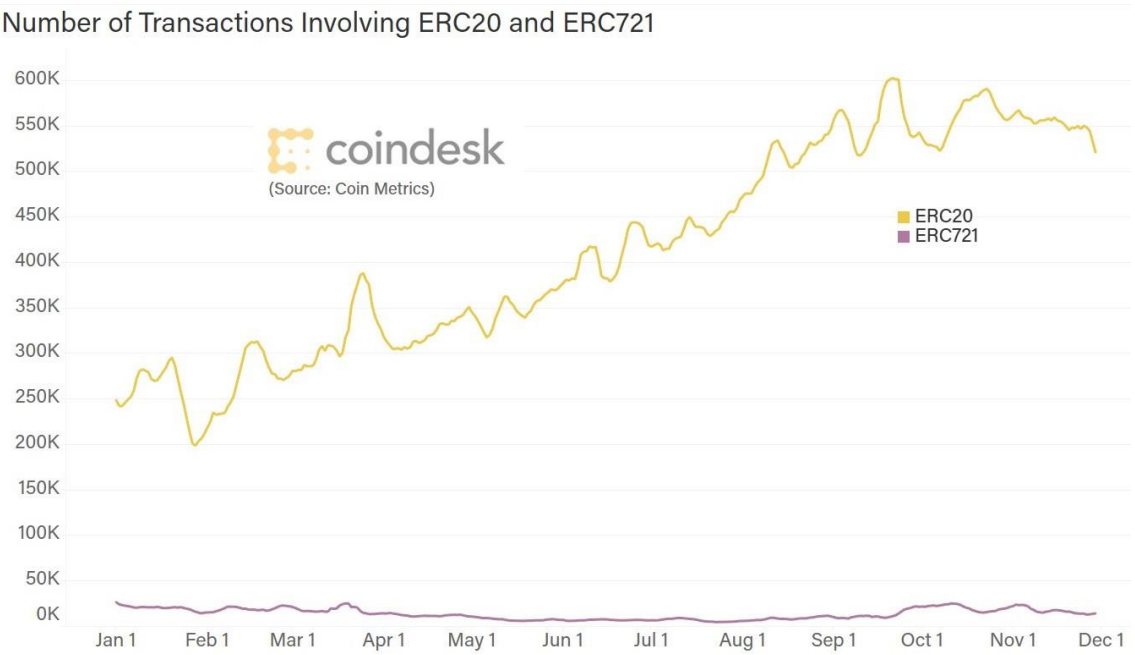
Of course, the growth in the number of ETH transacted (the number of units that moves between one address and another) is a more direct indicator of ETH demand, as it indicates the evolution of demand to buy and sell the asset as well as to open and fund dapps on the network.

Growing ETH demand



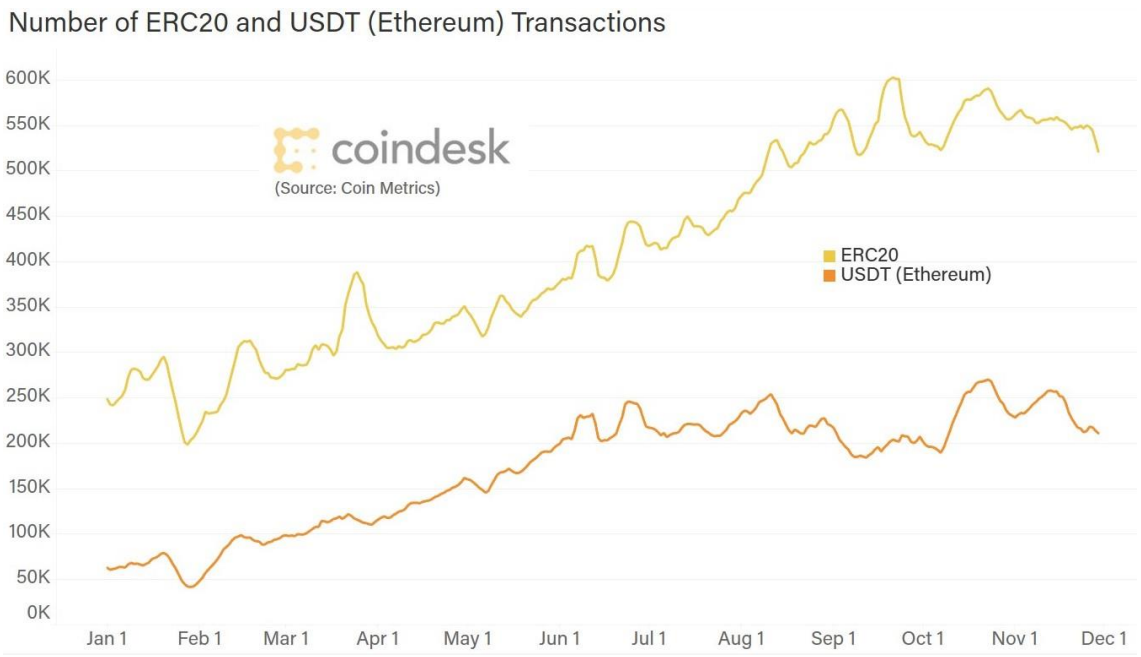
Ethereum transaction data also gives us a glimpse at the growth in usage of other types of tokens, such as those built on the ERC20 and ERC721 standards.

Evolution of ERC20 and ERC721 demand



The above chart shows that non-ETH activity on Ethereum is dominated by ERC20 tokens. The largest of these is tether (USDT), which accounts for about 40% of ERC20 transactions. (It is worth noting that USDT also runs on several other blockchains, so its transaction count on Ethereum is not an accurate representation of tether's overall growth.)

Tether on Ethereum is a major driver of ERC20 activity



CONCLUSION

Tracking transaction volumes on Ethereum is more complex than on other blockchains such as Bitcoin, since the information contained in transactions is much more varied in terms of inputs and outputs. Not all transactions transfer the native cryptocurrency ether, for instance – many contain snippets of code that represent instructions for smart contracts that keep record of other crypto assets such as USDT or LINK.

On the one hand, this makes it a bit harder to monitor growth in demand for ETH, the blockchain's native asset. On the other hand, it offers us a much richer array of information that lends insight into the growth of the ecosystem as a whole.

This note sets out an example of how understanding the data contained in public blockchain transactions, whatever the blockchain, can paint a nuanced picture of economic activity, and give much more detail to investment analysts than is generally available in traditional markets.

In summary, zeroing in on Ethereum transactions reveals important trends about how the network is evolving. They also show how looking at various layers of information about a transaction can add detail to our understanding of Ethereum's use case and value over time.

As smart contract functionality continues to develop, and as new tokens find utility, the story that this metric tells us will continue to reflect ever greater complexity and potential.

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